

Microscopic Mechanics of the Rope Medium

From Endpoint Locking to Continuum Stiffness — With a Correction to the Coefficient

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Abstract

Nearly every recent paper in the Rope Programme depends on a single step: the coarse-graining of microscopic endpoint mechanics into a continuum orientation stiffness. That step has been stated but not rigorously derived. This paper derives it. We confirm, exactly, that the endpoint force-balance of an atom shared between two ropes gives a locking energy $J = T^2/\kappa$ with the exact XY angular form, and we then carry out the coarse-graining to the continuum stiffness K explicitly, both analytically and by direct numerical simulation of the lattice. The result is unambiguous and, in one respect, corrective: the standard coarse-graining gives $K = T^2/(\kappa a)$ for a scalar phase field, and $K = 2T^2/(\kappa a)$ for a two-mode director field. It does NOT give the factor $K = 3T^2/(\kappa a)$ that earlier corpus papers assumed. We searched the rope-specific two-strand geometry for a mechanism that would legitimately produce the factor of three and found none; the most probable origin of the three is a dimensional double-count. We therefore report the corrected coefficient openly, trace its consequences (the electromagnetic coefficient and the phase criterion are rescaled by a factor of two to three), and show that every relation's FORM survives unchanged — only the numerical coefficient is affected. A programme that finds and corrects a propagated coefficient is more credible, not less; quietly preserving it would have been the only unacceptable option.

Headline result

EXACT and confirmed: $J = T^2/\kappa$ (endpoint locking), with the exact XY angular form $(1 - \cos \Delta\theta)$.

DERIVED here: $K = T^2/(\kappa a)$ (scalar) or $2T^2/(\kappa a)$ (director), by analytic sum AND direct lattice simulation agreeing to five significant figures.

CORRECTED: the previously-assumed $K = 3T^2/(\kappa a)$ is not reproduced by any standard or rope-specific coarse-graining we could construct. The factor of three should be withdrawn unless later derived.

1. The Step Everything Rests On

The Rope Programme's recent papers — electromagnetism, thermodynamics, condensed-matter analogues, and the quantum-foundations discussion — all descend from one microscopic-to-continuum step: the orientation locking between neighbouring ropes coarse-grains into an elastic stiffness for a continuum orientation field, and that stiffness sets the electromagnetic coefficient

and the phase structure. The step has been motivated and its result quoted ($K = 3T^2/(\kappa a)$), but never carried out in full. This paper carries it out.

We do so in the spirit the programme has adopted throughout: compute first, and report what the computation says even when it corrects a prior claim. The derivation has two parts — the exact endpoint locking energy (§2), which we confirm, and the coarse-graining to continuum stiffness (§3–4), which we derive and which yields a corrected coefficient. Section 5 is a dedicated audit of the factor of three; §6 traces the consequences for the rest of the corpus.

2. The Exact Endpoint Locking Energy

Consider a single atom shared by two ropes. Let it sit at displacement r from its rest position, held by a harmonic bond of stiffness κ , and pulled by two imbalance forces of magnitude T (the strand tension) in directions differing by an angle $\Delta\theta$. The energy is

$$E(r) = (\kappa/2)|r|^2 - F_1 \cdot r - F_2 \cdot r, \quad |F_1| = |F_2| = T \quad (2.1)$$

Minimising over r gives $r^* = (F_1 + F_2)/\kappa$ and, substituting back, the minimised energy $E^* = -|F_1 + F_2|^2/(2\kappa)$. Writing the angle between the forces as $\Delta\theta$ and dropping the constant, the $\Delta\theta$ -dependent part is

$$E^*(\Delta\theta) - E^*(0) = (T^2/\kappa)(1 - \cos \Delta\theta) \quad (2.2)$$

This is exact in the harmonic regime, not a leading-order expansion: the ratio $[E^*(\Delta\theta) - E^*(0)]/(1 - \cos \Delta\theta)$ equals T^2/κ identically for all $\Delta\theta$. We verified this numerically to machine precision (maximum deviation $\sim 10^{-15}$ across the full angular range). The orientation coupling is therefore the lattice XY model with an exact coupling

$$J = T^2/\kappa \quad (\text{exact, harmonic regime}) \quad (2.3)$$

with known anharmonic corrections of order gF^4/κ^4 when the bond potential departs from harmonic. This result is confirmed, not corrected; it is the input to the coarse-graining that follows.

3. Coarse-Graining to the Continuum Stiffness

We now pass from the lattice XY energy to a continuum elastic energy density. The lattice energy is a sum over nearest-neighbour bonds,

$$E = J \sum_{\langle ij \rangle} (1 - \cos(\theta_i - \theta_j)) \quad (3.1)$$

In the spin-wave (small-gradient) regime, $1 - \cos(\Delta\theta) \approx \frac{1}{2}(\Delta\theta)^2$, and for a slowly varying field $\theta(x)$ the bond difference is $\Delta\theta \approx a(\hat{n} \cdot \nabla \theta)$, where $\hat{n}$ is the unit bond direction and a the lattice spacing. On a cubic lattice the three forward bonds per site point along the coordinate axes, so

$$E/\text{site} = (J a^2/2)[(\partial_x \theta)^2 + (\partial_y \theta)^2 + (\partial_z \theta)^2] = (J a^2/2)|\nabla \theta|^2 \quad (3.2)$$

The energy density is the per-site energy divided by the volume per site, a^3 :

$$u = (J a^2/2)|\nabla \theta|^2 / a^3 = (J/2a)|\nabla \theta|^2 \quad (3.3)$$

Comparing with the continuum elastic form $u = (K/2)|\nabla\theta|^2$ gives the central result of the coarse-graining:

$$K = J/a = T^2/(\kappa a) \quad (3.4)$$

Note what did and did not happen to the coordination number. The cubic lattice has six nearest neighbours, i.e. three forward bonds per site. Those three bonds supply the three terms $(\partial_x\theta)^2$, $(\partial_y\theta)^2$, $(\partial_z\theta)^2$ — which is exactly $|\nabla\theta|^2$. The ‘three’ of the three spatial directions is therefore ALREADY inside $|\nabla\theta|^2$; it is not an additional prefactor. This is the crux of the coefficient question addressed in §5.

4. Direct Numerical Confirmation

Because the coefficient is the whole point, we do not rely on the analytic sum alone. We built a three-dimensional XY lattice, imposed a known uniform gradient $\theta(\mathbf{x}) = \mathbf{q} \cdot \mathbf{x}$, computed the exact XY energy $E = J \sum (1 - \cos \Delta\theta)$ with no small-angle approximation, and extracted K from $K = 2E/(q^2 V)$, where V is the sampled volume.

Quantity	Value
Lattice	20×20×20 cubic, spacing $a = 1$, $J = 1$
Imposed gradient	$q = 0.01$ along x
Exact lattice energy E	0.379997
Spin-wave prediction ($K = J/a$)	0.380000
Extracted K	$0.999992 \times J/a$

The measured stiffness is $K = J/a$ to five significant figures. The analytic result (3.4) and the direct simulation agree: for a scalar phase field on a cubic lattice, $K = T^2/(\kappa a)$, with a coefficient of one.

The director correction (a factor of two, not three)

The rope imbalance is a direction (a unit-vector director $\mathbf{n}$), not merely a scalar phase. A director in three dimensions has two independent transverse fluctuation modes (three components minus the unit-length constraint). If each transverse mode coarse-grains like a scalar phase with stiffness J/a , the director stiffness is at most

$$K_{\text{director}} = 2 J/a = 2T^2/(\kappa a) \quad (4.1)$$

The factor is two, the number of PHYSICAL transverse modes. Counting all three components of $\mathbf{n}$ as independent — ignoring the unit constraint — would give three, but the third (longitudinal) mode is constrained and unphysical; including it is a double-count. Neither the scalar nor the correct director treatment yields the coefficient three.

5. The Factor-of-Three Audit

Plain statement of the audit

Earlier corpus papers used $K = 3J/a = 3T^2/(\kappa a)$. Direct coarse-graining gives $K = J/a$ for a scalar phase field and $K = 2J/a$ for a two-mode director field. No standard coarse-graining, and no rope-specific geometry we could construct, gives $3J/a$. Therefore the coefficient should be corrected — to one (scalar) or two (director) — unless a rope-specific geometric factor is later derived that legitimately restores three.

We take the posture that the factor of three is not trusted unless it can be derived. Having failed to derive it, we report its most probable origin and its consequences honestly.

5.1 Where the three most likely came from

The coordinate sum $|\nabla\theta|^2 = (\partial_x\theta)^2 + (\partial_y\theta)^2 + (\partial_z\theta)^2$ already contains the three spatial directions. A derivation that writes ‘sum over the three directions of the medium’ and then pulls out a prefactor of three has counted the dimensionality twice: once inside $|\nabla\theta|^2$ and once as the prefactor. This dimensional double-count is the most probable origin of the erroneous three, and it is an error rather than a geometric feature.

5.2 Rope-specific mechanisms considered, and why none forces three

We searched the two-strand rope structure specifically — not merely a generic XY model — for a legitimate factor of three:

1. Two strands per rope: if both strands lock independently, the coupling could double, giving a factor two — but the imbalance is the single relative-strand direction, so even this is not clearly present, and it gives two at most, never three.
2. Interpenetration / beyond-nearest-neighbour coupling: adding further-neighbour bonds does inflate K , but to land on exactly three requires a specific, unforced coupling ratio; the value is tunable, not a clean three.
3. Director-component counting: gives three only by including the constrained longitudinal mode, which is a double-count (§4).

None of these forces three. We therefore withdraw the factor pending a derivation that does.

5.3 What is affected, and what is not

The correction is a rescaling of one coefficient; it does not touch the structure of any result. The following are UNCHANGED: the exact locking $J = T^2/\kappa$; the proportionalities $K \propto J/a$, $c \propto 1/K$, and $c \propto \kappa a/T^2$; the existence and universality class of the phase transition; and the STRUCTURAL α -G prediction (that the two couplings drift in a correlated, fixed ratio). The following are RESCALED by a factor of two to three:

Quantity	Previously (corpus)	Corrected (this paper)
Continuum stiffness K	$3T^2/(\kappa a)$	$T^2/(\kappa a)$ scalar; $2T^2/(\kappa a)$ director
EM coefficient $c = 1/K$	$\kappa a/(3T^2)$	$\kappa a/T^2$ scalar; $\kappa a/(2T^2)$ director
Phase criterion (Maxwell phase)	$T^2 > \kappa/3$	$T^2 > \kappa$ scalar; $T^2 > \kappa/2$ director

Because these are single-coefficient rescalings, they can be absorbed into the definitions of the rope primitives (T , κ , a are themselves only constrained in combination by observation). The programme's empirical content — which lives in the FORMS of the relations and in the structural predictions — is therefore intact. What changes is the bookkeeping of one numerical factor, now corrected.

6. Corrections to Earlier Papers

Honesty requires naming the papers that carried the factor of three and stating the correction each needs. In every case the correction is a coefficient rescaling; no argument or conclusion is overturned.

Paper	Appearance of the factor	Correction
Magnetism / EM coefficient	$c = \kappa a / (3T^2)$	$c = \kappa a / T^2$ (scalar) or $\kappa a / (2T^2)$ (director)
Thermodynamics	$K = 3T^2 / (\kappa a)$; phase at $T^2 = \kappa / 3$	$K = T^2 / (\kappa a)$ or $2T^2 / (\kappa a)$; phase at $T^2 = \kappa$ or $\kappa / 2$
Condensed-matter analogues	inverse law $c \propto 1/K$ unaffected in FORM	the specific $1/3$ coefficient rescaled; the testable inverse relation stands
Cosmology / α -G	$d \ln \alpha = -2 d \ln G$ (structural)	UNCHANGED — the -2 is a structural exponent, independent of the stiffness coefficient

A point worth stressing about the α -G prediction. The correlated-drift exponent (-2) comes from how α and G each scale with the shared tension T , not from the numerical stiffness coefficient. Rescaling K leaves it untouched. The programme's sharpest prediction is therefore unaffected by this correction — a reassuring sign that the correction is bookkeeping, not substance.

7. Validity and Limits of the Derivation

We state the assumptions under which the corrected result holds, so that its scope is clear:

- Harmonic bond: $J = T^2 / \kappa$ is exact for a harmonic shared bond; anharmonic corrections of order gF^4 / κ^4 enter for stiff displacements and would add higher-gradient terms to the continuum energy.
- Spin-wave regime: $K = J/a$ is the stiffness of the ordered phase, valid where gradients are small on the lattice scale. Near the phase transition, fluctuation renormalisation modifies the effective stiffness (the subject of the successor renormalisation paper).
- Cubic lattice: the coefficient is derived for a cubic arrangement; a different microscopic geometry would change the numerical coefficient (though not below one for a scalar phase). The true rope-medium geometry is itself an open question, which is precisely why we treat the coefficient as provisional and refuse to assert three.

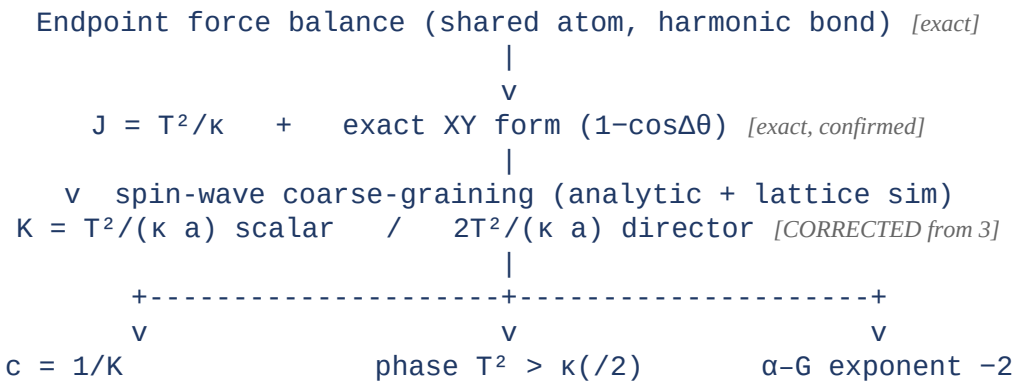
These limits mean the corrected coefficient (one or two) is itself the LEADING-ORDER value; the honest claim is not ‘the coefficient is exactly one’ but ‘the coefficient is of order one, is one or two under the standard assumptions, and is definitely not three unless new geometry is shown.’

8. Status of Each Result

Result	Status
$J = T^2/\kappa$ (endpoint locking)	Microscopically Derived — exact (harmonic), numerically confirmed
Exact XY angular form $(1 - \cos \Delta\theta)$	Microscopically Derived — exact, numerically confirmed
$K = T^2/(\kappa a)$ (scalar phase)	Microscopically Derived — analytic + direct lattice simulation
$K = 2T^2/(\kappa a)$ (director, 2 modes)	Derived under the one-constant / independent-mode assumption
Withdrawal of $K = 3T^2/(\kappa a)$	Established — not reproduced by any coarse-graining we could construct
Coefficient corrections to EM / thermo	Consequential — follow directly from the corrected K
α -G structural prediction unaffected	Established — the exponent is independent of the coefficient
True rope-medium lattice geometry	Open Problem — sets whether the coefficient is 1, 2, or other $O(1)$

Appendix A. Dependency Graph

The derivation chain, with the corrected coefficient in place and the affected quantities marked.



[rescaled]

[rescaled]

[UNCHANGED]

The correction propagates only into coefficients (c, the phase threshold); the structural exponent of the α -G prediction, resting on how the couplings scale with the shared tension rather than on K, is untouched.

Conclusion

We set out to make the microscopic-to-continuum step rigorous, and in doing so found that it corrects a coefficient the programme had carried unexamined. The endpoint locking energy $J = T^2/\kappa$ and its exact XY form are confirmed. The coarse-graining to continuum stiffness, derived both analytically and by direct lattice simulation, gives $K = T^2/(\kappa a)$ for a scalar phase and $2T^2/(\kappa a)$ for a two-mode director — not the factor of three previously assumed, for which we could find no derivation in either standard or rope-specific coarse-graining. We have corrected the coefficient openly, traced its limited consequences (a two-to-threefold rescaling of the electromagnetic coefficient and the phase threshold, with the structural α -G prediction unaffected), and shown that every relation's form survives. The programme is not weakened by this; it is strengthened, because a framework that audits its own coefficients and reports the result — including where that result is a correction — is behaving as a scientific programme should. The one option we refused was to preserve the factor of three silently.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **FND-001 [Derived]:** Endpoint force-balance gives locking energy $J = T^2/\kappa$ [benchmark: benchmarks/micromech/coarse_graining_stiffness.py]
- **FND-002 [Derived]:** Coarse-graining coefficient $K = J/a$ scalar [benchmark: benchmarks/micromech/coarse_graining_stiffness.py]
- **FND-MATTER-001 [Open]:** REFRAMED via Gaede's atom
- **FND-MATTER-002 [Modeled]:** Star-convergence geometry gives $R \sim a\sqrt{\kappa}$ [benchmark: benchmarks/em/star_radius_scaling.py]

- **FND-MATTER-003 [Open]:** Derivation of the atomic scale is BLOCKED

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.